

DC voltage on the output of the regulator 7905 should be around -5V, which depends to some extent on the value of VR1. The maximum output current of 7905 can be up to 1A and the maximum power dissipation is up to 15W. IC 7905 has internal

thermal protection.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Fix the stereo female jack on the front panel and speaker to the rear side of the cabinet, and the 12V battery inside the cabinet. Fix LED1

and switches S1 and S2 too on the front panel of the cabinet. Mount the regulator IC 7905 on a heat-sink with thermal resistance below 15°C/W. The metallic part on the case is internally connected with the input pin of the regulator. ●

Audible Logic Probe

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Using a conventional logic probe, it is difficult to determine what is happening at a point in a circuit above 50 Hz. The human eye isn't able to perceive the change which is happening at such a fast speed—for instance, the flickering of a tubelight.

The audible logic probe described here helps to assess the logic state at a

point in a circuit. The rate of change of state (frequency) is indicated by a tone. The audible probe has a large bandwidth ranging from DC to greater than 2 MHz. It uses a frequency-divider counter (CD4040), which divides the original signal frequency by 2^0 (1) to 2^{12} (4096).

The circuit (shown in Fig. 1) is built around op-amp LM339 (IC1) and CMOS chips CD4069 (IC2) and CD4040 (IC3). CD4040 is a 12-stage ripple-carry

binary counter/divider and oscillator. The tone is generated with the help of a piezobuzzer together with the divider counter. The piezobuzzer is driven by gates N3 and N5 of IC2, which are wired in anti-phase to each other. The probe can handle both TTL and CMOS-level supply.

The logic probe comprises the op-amp (IC1) and gates N1 and N2 of IC2,

which drive the dual-colour LED (LED1).

Three 10-kilo-ohm resistors (R3, R4 and R5) in series provide the biasing for the two op-amps in IC1. High logic is set at two-third of the supply voltage, whereas low logic is set at one-third of the supply voltage.

When the voltage at the probe goes below one-third of the supply voltage, the output of upper op-

amp/comparator IC1(A) goes low. When the probe voltage goes above two-third of the supply voltage, the output of lower comparator IC1(B) goes low, thus lighting the respective LEDs. When the red LED glows, the logic is low or '0'. When the green LED glows, the logic is high or '1'.

When the probe is placed at a particular frequency junction, the received clock is divided by CD4040 (IC3) to give an audio tone through

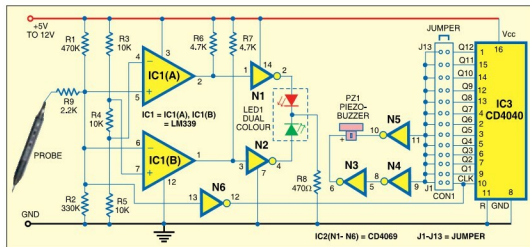


Fig. 1: Circuit for audible logic probe

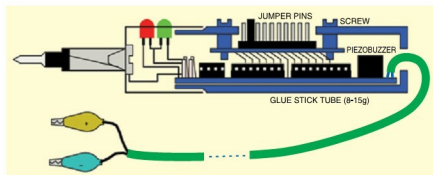


Fig. 2: Proposed assembly of logic probe